

Saugat Bhattarai

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EDUCATION

Masters in Data-driven Health

2025 – Present

KTH Royal Institute of Technology

Stockholm, Sweden

BE in Biomedical Engineering

College of Biomedical Engineering and Applied Sciences

Lalitpur, Nepal

2018 – 2023

- **CGPA:** 3.55/4.0
- **Relevant Coursework:** Biomedical Instrumentation, Medical Electronics Medical Image Processing, Neural Network, Biomedical Digital Signal Processing, Numerical Methods, Probability and Statistics, Calculus, Linear Algebra, Medical Imaging, Control Systems, Communication Systems, Electronic Devices and Circuits, Microprocessors, Fluid mechanics

RESEARCH EXPERIENCE

AI Fellowship Project

April 2024 – October 2024

Project Title: Brain Tumor Segmentation with synthetic MR Images

- Preprocessed MRI data from the BraTS2021 dataset, implementing standardized normalization and data augmentation protocols
- Developed and optimized baseline segmentation model and data pipeline, achieving 0.72 mean Dice Score for tumor detection

Bachelor Thesis

August 2022 – September 2023

Project Title: Treadmill-based Gait Analysis using Camera Kinematic System, Force Sensitive Resistors (FSR) and Motion Processing Unit (MPU) sensors

- Engineered an integrated Gait Analysis system using MPU, FSR and marker tracking, leveraging Python OpenCV and Arduino microcontroller
- Built data acquisition and preprocessing pipeline to extract joint angles and ground reaction forces
- Conducted comparative gait analysis between normal and abnormal subjects, validating system detection capabilities

WORK EXPERIENCE

National Electro-Tech Pvt. Ltd.

November 2023 – January 2024

- Installed, maintained and optimized ventilators, ultrasound machines, patient monitors, and plasma sterilizers
- Delivered training to healthcare professionals on proper device operation and preventive maintenance
- Executed routine servicing to ensure consistent reliability of medical equipment

ACADEMIC PROJECT

- **Scalable HPC-based Preprocessing of Liver Cancer Whole-Slide Images**
- **Arduino based Posture Detection system**
- **Simulation of an Artificial Cardiac Pacemaker using Proportional Integral Derivative (PID)**

INVOLVEMENT

AI Fellow

Fusemachines

April 2024 – October 2024

- Mastered Machine Learning and Computer Vision foundations, integrating core mathematical and data science principles
- Engineered Deep Learning architectures optimizing designs for practical applications

Trainee

AI in Healthcare

December 2024

- Engaged in workshop jointly organized by NepAI Applied Mathematics and Informatics Institute for research (NAAMII) & B.P. Eye Foundation, focusing on AI's role in Nepal's healthcare sector
- Covered AI/ML foundations, medical data acquisition, annotation challenges, ethical considerations, and regulatory aspects
- Engaged in discussions on AI model deployment, evaluation challenges, and gaps in ML research methodologies

Deerwalk Training Center

February 2024 – May 2024

- Built advanced proficiency in Python and essential Data Science libraries (NumPy, Pandas)
- Implemented Machine Learning algorithms, preprocessing pipelines, and foundational Deep Learning models
- Automated file handling, data manipulation and pattern recognition

United Nations Development Programme (UNDP) and Mission Oxygen Team (MOT)

August 2022

- Completed a five-day onsite training entitled “Strengthening Health System through trained Biomedical Engineers”
- Executed calibration, validation, and operation of critical medical equipment including X-ray, C-arm, infusion pumps and autoclaves
- Acquired expertise in medical equipment handling, sterilization procedures, and implementing safety protocols in healthcare settings

CERTIFICATIONS

- Unsupervised Learning, Recommenders, Reinforcement Learning, *Stanford University, DeepLearning.AI* • 2025
- Microdegree in Artificial Intelligence, *Fusemachines* • 2024
- AI for Medical Diagnosis, *DeepLearning.AI* • 2024
- Supervised Machine Learning: Regression and Classification, *Stanford University, DeepLearning.AI* • 2024
- Advanced Learning Algorithms, *Stanford University, DeepLearning.AI* • 2024
- Data Science and Machine Learning in Python, *Deerwalk Training Center* • 2024
- Machine Learning with Python, *IBM* • 2023
- Introduction to Data Science, *Cisco Networking Academy* • 2023
- Data Analysis with Python, *IBM* • 2023

SKILLS

- Programming and ML Frameworks: Python, MATLAB, SQL, R; TensorFlow, PyTorch, Scikit-learn
- Applied AI: Machine Learning, Deep Learning, Computer Vision
- Data Engineering and Pipelines: Data Preprocessing, Feature Engineering, Environment setup
- Data Analysis and Visualization: Pandas, NumPy, Matplotlib, Seaborn, OpenCV
- Version Control and Collaboration: Git, GitHub
- IoT for Health Applications: Arduino, PCB Design, Sensor integration, Health data acquisition and processing

PROFESSIONAL AFFILIATION

- **Nepal Engineering Council**
Certified Registered Engineer in Biomedical Engineering
License Number: 83961